RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)
Department of Artificial Intelligence | V Semester B.Tech AIDS

AD23532 – Principles of Data Science

Ex1: Data Preprocessing and Cleaning

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Task 1 – Import the required Python libraries

Libraries imported (numpy, pandas, matplotlib, seaborn) and Telco Customer Churn dataset loaded into a DataFrame.

```
[ ]  import numpy as np
     import pandas as pd
     import matplotlib.pyplot as plt
     import seaborn as sns

[ ]  df=pd.read_csv(r"/content/WA_Fn-UseC_-Telco-Customer-Churn.csv")
```

Task 2 – Load the dataset: first 10 records, last 10 records, shape, column names

First 10 records – `df.head(10)`

```
[ ]  df.head(10)
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	Pay
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	No	No	Month-to-month	Yes	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	No	No	One year	No	M
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	No	No	Month-to-month	Yes	M
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No	No	One year	No	B
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	No	No	Month-to-month	Yes	
5	9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	...	Yes	No	Yes	Yes	Month-to-month	Yes	
6	1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	...	No	No	Yes	No	Month-to-month	Yes	
7	6713-OKOMC	Female	0	No	No	10	No	No phone service	DSL	Yes	...	No	No	No	No	Month-to-month	No	M
8	7892-POOKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	...	Yes	Yes	Yes	Yes	Month-to-month	Yes	
9	6388-TABGU	Male	0	No	Yes	62	Yes	No	DSL	Yes	...	No	No	No	No	One year	No	B

Last 10 records and shape of the dataset – `df.tail(10)`, `np.shape(df)` = *(7043, 21)*

```
[ ]  df.tail(10)
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling
7033	9767-FFLEM	Male	0	No	No	38	Yes	No	Fiber optic	No	...	No	No	No	No	Month-to-month	Yes
7034	0639-TSIQW	Female	0	No	No	67	Yes	Yes	Fiber optic	Yes	...	Yes	No	Yes	No	Month-to-month	Yes
7035	8456-QDAVC	Male	0	No	No	19	Yes	No	Fiber optic	No	...	No	No	Yes	No	Month-to-month	Yes
7036	7750-EYXWZ	Female	0	No	No	12	No	No phone service	DSL	No	...	Yes	Yes	Yes	Yes	One year	No
7037	2569-WGERO	Female	0	No	No	72	Yes	No	No	No internet service	...	No internet service	No internet service	No internet service	No internet service	Two year	Yes
7038	6840-RESVB	Male	0	Yes	Yes	24	Yes	Yes	DSL	Yes	...	Yes	Yes	Yes	Yes	One year	Yes
7039	2234-XADUH	Female	0	Yes	Yes	72	Yes	Yes	Fiber optic	No	...	Yes	No	Yes	Yes	One year	Yes
7040	4801-JZAZL	Female	0	Yes	Yes	11	No	No phone service	DSL	Yes	...	No	No	No	No	Month-to-month	Yes
7041	8361-LTMKD	Male	1	Yes	No	4	Yes	Yes	Fiber optic	No	...	No	No	No	No	Month-to-month	Yes
7042	3186-AJIEK	Male	0	No	No	66	Yes	No	Fiber optic	Yes	...	Yes	Yes	Yes	Yes	Two year	Yes

10 rows × 21 columns

```
[ ]  np.shape(df)

     (7043, 21)
```

Column names – `df.columns`

```
[ ]  df.columns

     Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
            'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
            'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
            'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
            'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
           dtype='object')
```

Task 3 – Explore the dataset

Dataset information – `df.info()`

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

Summary statistics – `df.describe()`

```
df.describe()
```

	SeniorCitizen	tenure	MonthlyCharges
count	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692
std	0.368612	24.559481	30.090047
min	0.000000	0.000000	18.250000
25%	0.000000	9.000000	35.500000
50%	0.000000	29.000000	70.350000
75%	0.000000	55.000000	89.850000
max	1.000000	72.000000	118.750000

Number of missing values – `df.isnull().sum()`

```
df.isnull().sum()
```

	0
customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0

Number of duplicate records – `df.duplicated().sum()` = *0*

```
df.duplicated().sum()

np.int64(0)
```

Task 4 – Clean the data

4a. Remove duplicate customer records – `df.drop_duplicates()`

```
df.drop_duplicates()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	No	No	Month-to-month	Yes
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	No	No	One year	No
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	No	No	Month-to-month	Yes
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No	No	One year	No
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	No	No	Month-to-month	Yes
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
7038	6840-RESVB	Male	0	Yes	Yes	24	Yes	Yes	DSL	Yes	...	Yes	Yes	Yes	Yes	One year	Yes
7039	2234-XADUH	Female	0	Yes	Yes	72	Yes	Yes	Fiber optic	No	...	Yes	No	Yes	Yes	One year	Yes
7040	4801-JZAZL	Female	0	Yes	Yes	11	No	No phone service	DSL	Yes	...	No	No	No	No	Month-to-month	Yes
7041	8361-LTMKD	Male	1	Yes	No	4	Yes	Yes	Fiber optic	No	...	No	No	No	No	Month-to-month	Yes
7042	3186-AJIEK	Male	0	No	No	66	Yes	No	Fiber optic	Yes	...	Yes	Yes	Yes	Yes	Two year	Yes

7043 rows × 21 columns

4b/4c. Replace missing values (numerical → median, categorical → mode) and verify with `df.isnull().sum()`

```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')

numerical_cols = df.select_dtypes(include=np.number).columns
categorical_cols = df.select_dtypes(include='object').columns

for col in numerical_cols:
    if df[col].isnull().any():
        median_val = df[col].median()
        df[col].fillna(median_val, inplace=True)
        print(f"Filled missing values in numerical column '{col}' with median: {median_val}")

for col in categorical_cols:
    if df[col].isnull().any():
        mode_val = df[col].mode()[0]
        df[col].fillna(mode_val, inplace=True)
        print(f"Filled missing values in categorical column '{col}' with mode: {mode_val}")
```

```
Filled missing values in numerical column 'TotalCharges' with median: 1397.475
/tmp/ipykernel_1301/2354683027.py:9: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.


  df[col].fillna(median_val, inplace=True)
```

```
df.isnull().sum()
```

	0
customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0

Task 5 – Encode categorical columns

One-hot encoding using `pd.get_dummies(..., drop_first=True)`*; new shape (7043, 32)*

```
columns_to_encode = ['gender', 'Partner', 'Dependents', 'Contract', 'PaymentMethod', 'InternetService', 'Churn', 'PhoneService', 'MultipleLines', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', '
df_encoded = pd.get_dummies(df, columns=columns_to_encode, drop_first=True)
print("DataFrame after one-hot encoding:")
display(df_encoded.head())
print(f"New shape of the DataFrame: {df_encoded.shape}")
```

DataFrame after one-hot encoding:

	customerID	SeniorCitizen	tenure	MonthlyCharges	TotalCharges	gender_Male	Partner_Yes	Dependents_Yes	Contract_One year	Contract_Two year	...	OnlineBackup_Yes	DeviceProtection_No internet service	DeviceProtection_Yes	TechSupport_No internet service	TechSu
0	7590-VHVEG	0	1	29.85	29.85	False	True	False	False	False	...	True	False	False	False	
1	5575-GNVDE	0	34	56.95	1889.50	True	False	False	True	False	...	False	False	True	False	
2	3668-QPYBK	0	2	53.85	108.15	True	False	False	False	False	...	True	False	False	False	
3	7795-CFOCW	0	45	42.30	1840.75	True	False	False	True	False	...	False	False	True	False	
4	9237-HQITU	0	2	70.70	151.65	False	False	False	False	False	...	False	False	False	False	

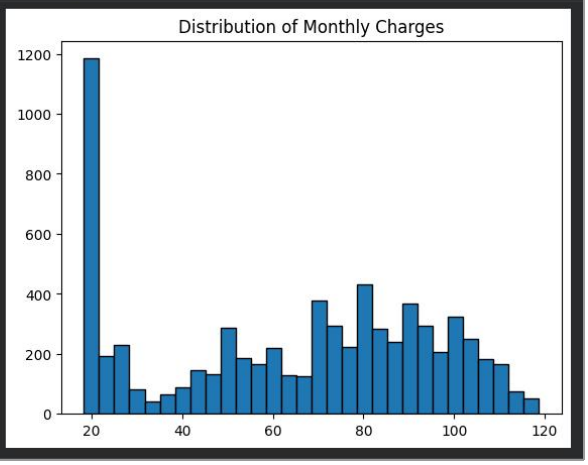
5 rows × 32 columns

New shape of the DataFrame: (7043, 32)

Task 7 – Plot distribution of MonthlyCharges and Tenure

Distribution of Monthly Charges

```
plt.hist(df['MonthlyCharges'], bins=30, edgecolor='black')
plt.title('Distribution of Monthly Charges')
plt.show()
```



Distribution of Tenure

```
plt.hist(df['tenure'], bins=30, edgecolor='black')
plt.title('Distribution of Tenure')
plt.show()
```

